Reduced-Form Analysis: Weather-Mortality Gradient on the CMR

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1 Research Question

Did the 2017 EU–Libya Memorandum of Understanding change how sea conditions affect migrant drowning deaths in the Central Mediterranean? Specifically, did weather deterrence — the pattern where rough seas reduce deaths because fewer people attempt crossing — weaken or reverse after the MoU?

2 Estimand

The coefficient β_3 in:

$$\mathbb{E}[\text{deaths}_t] = \exp(\beta_1 \cdot \text{SWH}_t + \beta_3 \cdot \text{SWH}_t \times \text{post\_mou} + \alpha_{\text{month-year}})$$

- β_1: SWH–mortality gradient in the pre-MoU period
- β_3: change in this gradient after the MoU
- $\exp(\beta_3)$ = IRR: multiplicative change in the SWH–deaths elasticity

Interpretation:

- $\beta_1 < 0$ (deterrence): rougher weather $\rightarrow$ fewer deaths (fewer crossings)
- $\beta_3 > 0$: deterrence weakened post-MoU
- $\beta_1 + \beta_3$: total gradient post-MoU (if positive: rough weather now increases deaths)

3 Data

3.1 Outcome: Drowning Deaths

Source: IOM Missing Migrants Project (2014+), Migrant Files (pre-2014)

Filters applied to IOM data:

- Route: Central Mediterranean only
- Incident Type: "Incident" only (excludes Cumulative and Split Incidents, which have imprecise dates and can double-count)
- Cause of death: "Drowning" or "Mixed or unknown" (bodies found at sea where cause was undetermined). Excludes sickness, violence, vehicle accidents — SWH does not mechanistically cause these.

- Geography: Core corridor $[10.0, 15.1] \times [32.4, 37.8]$ for primary specification (matches SWH extraction zone). All CMR as robustness.

Migrant Files (2008–2013):

- Researcher-compiled database from media reports and NGO sources
- Used for pre-IOM period only (before 2014-01-01); IOM takes precedence for all dates it covers
- Diagnostic (script 05c) shows MF data too sparse at daily level (127 non-zero days in 6 years, 5.8% event density) for gradient estimation. Overall gradient is null ($b = +0.173$, SE = 0.229, $p = 0.45$).
- MF used at weekly level in the weekly panel robustness (05e), where data density is viable

3.2 Weather: Significant Wave Height (SWH)

Source: ERA5 reanalysis (ECMWF), daily, 0.5-degree resolution

Extraction: Spatial mean over ocean cells within the `sea_zone_core` polygon, defined in `00_define_sea_zones.R`. The polygon follows Tunisian/Libyan coastlines, diagonal Cap Bon to W Sicily, Sicily south coast, east edge at 16E. Approximately $[10.0, 15.1] \times [32.4, 37.8]$.

Lag structure:

- Primary: `swh_prevweek` — mean SWH over days $t-1$ to $t-7$ (captures transit weather during the $ $1–7 day crossing)
- Robustness: `swh_prev3days` — mean SWH over days $t-1$ to $t-3$
- Falsification: `swh_next7avg` — mean SWH over days $t+1$ to $t+7$ (future weather cannot cause past deaths)

Standardization: All SWH variables z-scored (mean 0, SD 1) within each sample period so coefficients are per 1-SD and comparable across lag structures.

3.3 Additional Data (built into daily panel by 00b)

Variable	Frequency	Source
UNHCR daily arrivals to Italy	Daily	UNHCR
IOM official sea arrivals	Monthly	IOM MMP spreadsheet
LCG + TCG interceptions	Monthly	IOM MMP spreadsheet (2016+, NA before)
NGO SAR vessel count (24 tracked)	Daily	Rodriguez-Sanchez et al. (2023)
Government SAR operations	Daily	Rodriguez-Sanchez et al. (2023)

Note on the daily panel (`00b_build_daily_panel.R`): The panel spans 2008–2025 and includes: date, deaths, n_incidents, arrivals, intercept_per_day, official_arrivals, interceptions, swh, swh_prev3days, swh_prevweek, crossings, crossings_no_ic, ym, iso_week. Deaths come from Migrant Files (2008–2013) and IOM MMP (2014+) with no overlap. Arrivals are available from Oct 2015 (UNHCR). Interceptions are monthly totals spread to daily rate.

4 Descriptive Analysis Pipeline (Scripts 01–04)

4.1 Weather Danger Analysis (01)

Descriptive analysis of crossing danger by sea state, answering three questions:

- **Q1:** How dangerous is being at sea during different weather? Fatality rate (deaths/crossings) computed by sea state category (Calm < 0.5m, Medium 0.5–1.25m, Rough > 1.25m), using prev-3d SWH.
- **Q2:** Is being at sea during rough conditions more dangerous post-MoU? Same fatality rates split by pre/post-MoU period.
- **Q3:** Have crossings in rough conditions increased or decreased? Share of total crossings by sea state and period.

Computed at both daily and weekly frequency. Period: 2015–2021 (where arrivals data is available).

4.2 Fatality Rate Time Series (02)

Weekly and monthly fatality rate time series. Compares two definitions: deaths/(deaths+arrivals) vs deaths/(deaths+arrivals+interceptions). Monthly panel also overlays IOM's official CMR death rate.

4.3 Crossing Components (03)

Stacked area plots decomposing total crossings into arrivals, interceptions, and deaths, with fatality rate overlaid as LOESS-smoothed black line. Weekly (from Oct 2015) and monthly (from 2014) panels.

4.4 SWH vs Fatality Rate (04)

Camarena et al. (2020)-style dual-axis time series: LOWESS-smoothed death rate (bw $= 0.025$) + raw prior-week SWH. Three-panel comparison: (A) Camarena et al. replication data, (B) our data without interceptions, (C) our data with interceptions.

5 Primary Model Specification (Script 05)

5.1 Estimator

Distribution: Negative Binomial (`fixest::fenegbin`)

- NegBin chosen over Poisson due to extreme overdispersion (variance/mean $\gg 1$)
- Overdispersion from mass-casualty events (days with $>$100 deaths)

Fixed effects: Month-year ($ $96 cells for 2014–2021, $ $132 for 2014–2024)

- Absorbs seasonality AND year-specific level shifts in deaths
- Identification from day-to-day weather variation within months
- Justified over coarser alternatives (see Section 7.2)

Standard errors: Newey–West(28) primary, justified by SE diagnostic (Section 6)

- NW(14) and HC as robustness
- Cluster(iso_week) as additional robustness

Formula:

```
n_dead_missing ~ swh_prevweek_z + swh_prevweek_z:post_mou | month_year
```

The main effect of `post_mou` is absorbed by month-year FE (each month-year cell is entirely pre or post). A `unit = 1L` panel ID is created for the NW variance estimator (`panel.id = ~unit + date`).

5.2 Robustness Specifications

Spec	What varies	Purpose
Prev-3d SWH	Lag window	Shorter transit time assumption
All CMR geography	Incident geography	Tests sensitivity to corridor restriction
No outliers (deaths ≤ 100)	Outcome trimming	Tests if mass-casualty events drive result
Next-7d SWH (falsification)	Timing	Future weather cannot cause past deaths

5.3 Year-by-Year Gradient

No break date assumed. One SWH coefficient per year:

```
n_dead_missing ~ swh_prevweek_z:year_fac | month_year
```

Shows how the gradient evolves over time without imposing a pre/post structure. NW(28) SEs used for inference.

5.4 Sample Periods

All specifications run for two periods:

- **2014–2021:** Primary period
- **2014–2024:** Extended period (tests stability over longer horizon)

SWH is re-standardized within each sample period.

6 SE Diagnostic (Script 05f)

6.1 Motivation

The choice of standard errors matters substantially for inference. Script 05f diagnoses serial correlation in residuals and compares SE approaches.

6.2 Residual Autocorrelation

Pearson residuals from the primary NegBin model (2014–2024, $N = 4{,}018$) show:

- Individual autocorrelations are small (lag-1 $= -0.012$, lag-7 $= -0.020$)
- But **Breusch–Godfrey rejects no serial correlation at all orders** (OLS proxy):

- Order 7: LM = 48.38, $p < 0.001$
- Order 14: LM = 113.27, $p < 0.001$
- Order 28: LM = 312.11, $p < 0.001$

This means HC SEs are anti-conservative despite small individual autocorrelations. NW SEs are required.

6.3 SE Comparison (2014–2024, primary specification)

From `output/tables/se_diagnostic_comparison.txt`:

VCOV	β	SE	z	p
HC	+0.996	0.250	+3.98	0.0001
NW(7)	+0.996	0.351	+2.83	0.0046
NW(14)	+0.996	0.400	+2.49	0.0127
NW(28)	**+0.996**	**0.446**	**+2.23**	**0.0255**
Cluster(week)	+0.996	0.337	+2.96	0.0031

SE inflation relative to HC: NW(7) = 1.40×, NW(14) = 1.60×, **NW(28)** = 1.78×, Cluster = 1.35×.

Decision: NW(28) is primary because Breusch–Godfrey rejects at all orders up to 28, and NW(28) is the most conservative parametric option tested. The result remains significant ($p = 0.026$) but substantially less so than with HC ($p = 0.0001$).

7 Diagnostic Tests (Scripts 05b–05e)

7.1 Cutoff Sensitivity / Placebo Test (05b)

Question: Is the MoU date (July 2017) uniquely special?

Method: Re-estimate the interaction model at every possible monthly cutoff date within the sample period. Uses HC SEs for computational feasibility across all placebo runs.

Answer: No. Many dates produce $|\beta_3| \geq$ the MoU's $|\beta_3|$.

The gradient shift is a broad pattern across \$ \$2016–2019, not a sharp break at the MoU. Consistent with a gradual policy transition (LCG operations ramped up slowly; NGO pullback was incremental; Code of Conduct implemented in steps).

Also produces a NegBin model-implied gradient plot showing pre vs post SWH–deaths relationship with confidence bands.

7.2 FE Structure Diagnostic (05d)

Question: Why month-year FE rather than week-of-year (Deiana et al. 2024) or month-of-year (Zambiasi & Albarosa 2025)?

Three tests (all use HC SEs for within-diagnostic comparability):

Test 1: Post-MoU deaths dropped \$ \$60% at ALL SWH levels (low, medium, high terciles). This is a level shift, not a gradient shift. Coarser FE that don't absorb year-specific levels conflate the two.

Test 2: FE comparison (2014–2021, HC SEs):

FE	β_3	p
Month-of-year (Zambiasi)	+0.160	0.453
Week-of-year (Deiana)	+0.251	0.181
Month-of-year + Year	+0.451	0.048
Month-year (primary)	+1.241	$<$0.001

Adding year controls recovers the signal. Note: the $p < 0.001$ for month-year FE uses HC SEs; with NW(28), $p = 0.022$ (see Section 8).

Test 3: A single `post_mou` dummy (instead of year FE) is insufficient: $\beta_3 = +0.209$ ($p = 0.332$). Year-by-year variation in death levels cannot be captured by one binary indicator.

Conclusion: Coarser FE work for Deiana (who exploits SAR operation timing variation) and Zambiasi (who uses spatial distance variation). Our design uses only temporal weather variation, requiring month-year FE to absorb year-specific death levels unrelated to weather.

7.3 Migrant Files Diagnostic (05c)

Question: Can we extend the pre-period to 2008 using Migrant Files?

Answer: No, at daily level. From `output/tables/migrant_files_diagnostic.txt`:

- $N = 2{,}185$ days, only 127 with deaths (5.8%) vs IOM's 13.6%
- Overall gradient: $b = +0.173$ (SE = 0.229, $p = 0.45$) — null
- Year-by-year oscillates wildly: 2009 ($+1.77^*$), 2011 (-2.28^*), 2013 (-3.26^*)

Decision: Daily panel starts from 2014 (IOM). Weekly panel extends to 2011 for robustness.

7.4 Weekly Panel (05e)

Question: Does the result hold at weekly frequency?

Method: Same reduced-form design aggregated to weekly level. FE: month-of-year + year (since month-year has only $ $4 obs per cell at weekly frequency). Extends to 2011 using Migrant Files.

Answer: No — weekly panel gives null results. From `output/tables/weekly_panel_results.txt`:

Period	Spec	β_3	SE	p
2011–2021	Primary	+0.273	0.259	0.291
2011–2021	All CMR	+0.351	0.207	0.090
2011–2024	Primary	+0.252	0.213	0.236
2011–2024	All CMR	+0.344	0.175	0.049

Root cause: Weekly averaging smooths out high-frequency weather variation. FE absorb 46% of weekly SWH variation vs 23% of daily. After absorbing seasonality, insufficient residual SWH variation remains at weekly frequency.

This is a power issue, not evidence against the daily result. The daily model exploits day-to-day weather swings within months; these are averaged out at weekly level.

8 Key Results (NW(28) SEs)

8.1 Primary Estimates

From `output/tables/reduced_form_results.txt`:

2014–2021:

Spec	β_3	SE	IRR	p
Prev-week (primary)	+1.241	0.540	3.458	0.0215
Prev-3d (robustness)	+0.465	0.359	1.592	0.1956
All CMR (robustness)	+1.027	0.402	2.792	0.0106
No outliers (robustness)	+1.028	0.457	2.795	0.0244
Next-7d (falsification)	\$-\$0.234	0.456	0.791	0.6074

2014–2024:

Spec	β_3	SE	IRR	p
Prev-week (primary)	+0.996	0.446	2.707	0.0255
Prev-3d (robustness)	+0.342	0.262	1.407	0.1920
All CMR (robustness)	+0.755	0.332	2.128	0.0230
No outliers (robustness)	+0.806	0.348	2.239	0.0205
Next-7d (falsification)	+0.049	0.357	1.051	0.8901

8.2 Robustness Summary

- **Spatial definition:** Robust. All CMR (no corridor restriction) significant at $p < 0.025$ in both periods.
- **Outlier trimming:** Robust. Removing days with \$>\$100 deaths: $p = 0.024$ / 0.021.
- **Timing window:** Not robust. Prev-3d SWH is not significant ($p = 0.20$ / 0.19). The 3-day window is likely too short to capture crossing transit weather.
- **Falsification:** Passes. Future weather (next-7d) has no effect ($p = 0.61$ / 0.89).

8.3 Year-by-Year Gradient

From the year-by-year model. The gradient is not stable — it oscillates substantially over the full period (NW(28) SEs):

- **Negative (deterrence):** 2015 (-1.75), 2016 (-0.56), 2017 (-0.71), 2022 (-0.88^*)
- **Positive (danger):** 2018 (+0.75), 2021 (+0.76), 2023 (+0.67)
- Wide CIs for most years; few individually significant

The transition from deterrence to danger occurs around 2018, not at the MoU signing (Feb/July 2017). The effect is not permanent — it partially reverses in 2022 and oscillates after.

Note: Year-by-year coefficients are printed to console by script 05 but not saved to file. The numbers above are from CLAUDE.md, which recorded them from a prior run. They should be verified by re-running the script.

8.4 Interpretation

The reduced-form interaction β_3 is positive and significant: after the MoU, the SWH–mortality gradient shifted. Pre-MoU, the gradient was negative (rough weather associated with fewer deaths — deterrence). Post-MoU, the total gradient ($\beta_1 + \beta_3$) became positive (rough weather associated with more deaths — danger dominates).

However, the placebo test (Section 7.1) shows this date is not uniquely special — many cutoff dates produce similar or larger coefficients. The year-by-year gradient oscillates rather than showing a clean break. The result documents a real shift in the weather–mortality relationship, but cannot attribute it specifically to the MoU.

9 Limitations

1. **The MoU date is not uniquely identified.** The placebo test shows many dates give comparable coefficients. The gradient shift is gradual, not a sharp break. This limits causal attribution to the MoU specifically.
2. **The prev-3d timing alternative is not significant.** The result is robust to spatial definition and outlier trimming, but not to the shorter weather window ($p \approx 0.19$). This may reflect that 3 days is too short to capture the relevant transit weather window, rather than a fragility of the result.
3. **The reduced-form mixes channels.** We cannot separate whether the gradient shift is because (a) more people cross in rough weather (deterrence collapse) or
   (b) each rough-weather crossing became more dangerous. Both are consistent with $\beta_3 > 0$.
4. **Extreme overdispersion.** Driven by mass-casualty events. NegBin handles this, and the no-outliers robustness confirms the result isn't driven by the largest events.
5. **IOM reporting dates.** The IOM date is the reporting date, not the crossing date. Lag-1 to lag-7 SWH addresses this. Reporting coverage may have changed post-MoU as SAR patterns shifted.
6. **No crossing-level data.** Without individual crossing records (boat-level data), we cannot decompose the reduced-form into deterrence vs danger channels.
7. **HC vs NW SEs matter substantially.** HC SEs give $p < 0.001$; NW(28) gives $p = 0.026$. All primary results use NW(28), the most conservative option. This SE sensitivity should be transparently reported.

10 Output Files

10.1 Scripts

Script	Purpose
00_define_sea_zones.R	Define core corridor sea zone polygon
00b_build_daily_panel.R	Build daily panel (weather + deaths + arrivals + interceptions)
01_weather_danger_analysis.R	Fatality rate by sea state, pre/post MoU

Script	Purpose
`02_fatality_rate_timeseries.R`	Weekly and monthly fatality rate time series
`03_crossing_components.R`	Stacked crossing components with fatality rate overlay
`04_swh_vs_fatality_rate.R`	Camarena-style SWH vs death rate dual-axis plots
`05_reduced_form_model.R`	Primary model, both periods, all specs + year-by-year
`05b_reduced_form_plots.R`	Gradient plot + placebo sensitivity plot
`05c_migrant_files_diagnostic.R`	Why daily panel starts from 2014
`05d_fe_structure_diagnostic.R`	Why month-year FE is necessary
`05e_weekly_panel_model.R`	Weekly robustness + why it gives null
`05f_se_diagnostic.R`	Residual ACF, Breusch–Godfrey, NW vs HC comparison

10.2 Key Figures

File	Content
`reduced_form_coefplot.png`	Coefficient plot: β_3 across specs and periods
`reduced_form_yearly_gradient.png`	Year-by-year SWH coefficient trajectory
`reduced_form_gradient.png`	Model-implied gradient, pre vs post, with CIs
`reduced_form_placebo.png`	Cutoff sensitivity: β_3 at all placebo dates
`fe_structure_comparison.png`	FE structure comparison
`se_diagnostic_acf.png`	ACF and PACF of Pearson residuals

10.3 Key Tables

File	Content
`reduced_form_results.txt`	Full interaction results (NW(28) SEs)
`fe_structure_diagnostic.txt`	FE diagnostic results (HC SEs)
`se_diagnostic_comparison.txt`	SE comparison across VCOV types
`weekly_panel_results.txt`	Weekly panel results + variation loss explanation
`migrant_files_diagnostic.txt`	MF sparsity diagnostic

11 Relationship to Literature

Deiana, Maheshri, and Mastrobuoni (2024, AEJ:EP): Use daily Poisson QMLE with week-of-year FE to study how SAR affects the SWH–crossings elasticity. Our model adapts their weather interaction approach but uses NegBin (for overdispersion) and month-year FE (for year-specific level shifts). They study crossings; we study deaths. Their Corollary 1 (crossings become less elastic to weather under SAR) is the mechanism behind our finding.

Zambiasi and Albarosa (2025, JEG): Use spatial DiD (distance from Libya × post) with month + quadrant FE. They control for SWH as a covariate. We use SWH as the key independent variable. Their finding (deaths moved closer to Libya) complements ours (weather deterrence collapsed). They use month-of-year FE — our FE diagnostic shows why month-year FE is necessary for our design.

Camarena et al. (2020): Dual-axis plots of death rate and wave height. Script 04 replicates their visualization approach with our data. Our contribution is formalizing the SWH–mortality relationship in a regression framework rather than visual inspection.